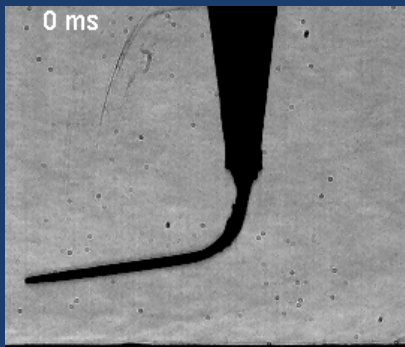


Traversing Pitot Probe Characterization of the UTSA Mach 7 Wind Tunnel

Nandan Manjith

UTSA Hypersonics Lab

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Motivation

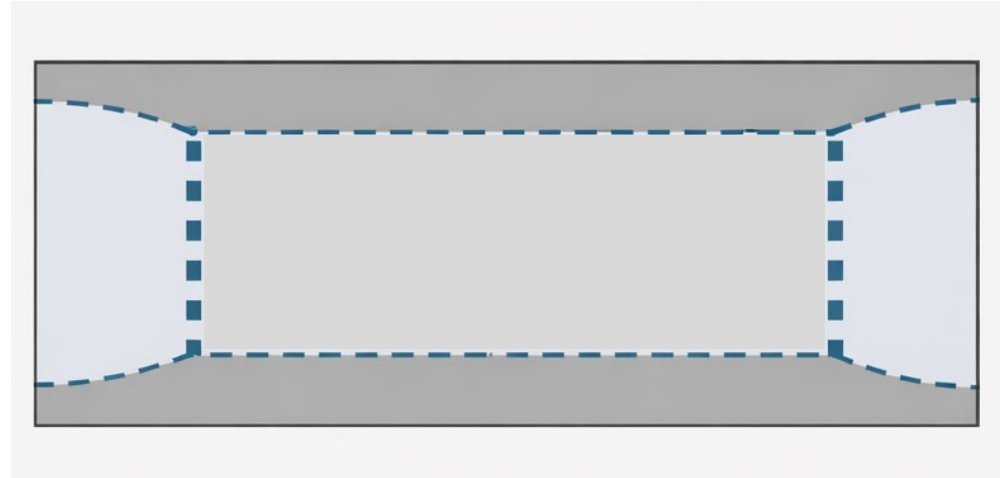
- Hypersonic flight ($\text{Mach} > 5$) is critical for atmospheric entry and defense systems
- Flow physics at hypersonic speeds are complex and difficult to predict
- Wind tunnel experiments provide controlled environments to study and validate these flows

Advancements in hypersonic flight depends on wind tunnel testing and accurate flow characterization.



Challenges in Hypersonic Wind Tunnels

- Planar nozzle generates boundary layers along all walls
- Freestream region depends on boundary layer thickness



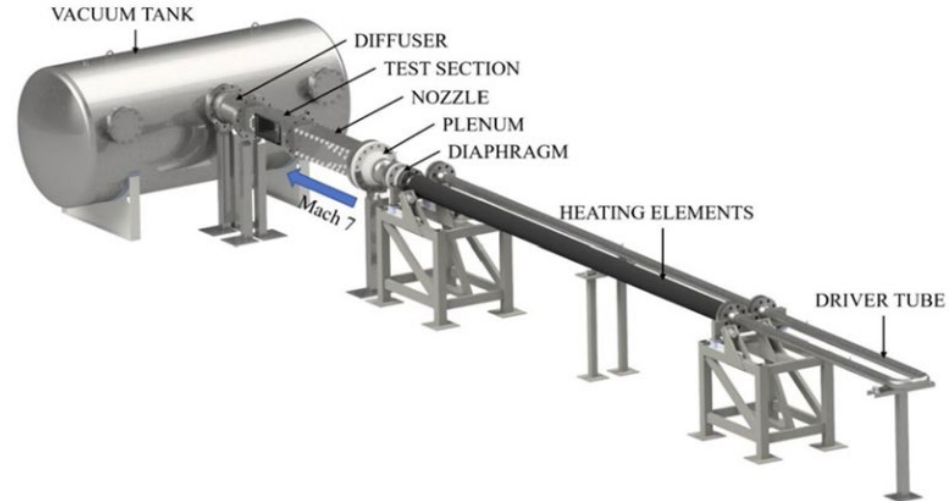
Planar Nozzle Cross Section

Research Objective

- Validate freestream Mach number of UTSA Mach 7 facility
- Measure boundary layer development
- Compare results with prior characterization data

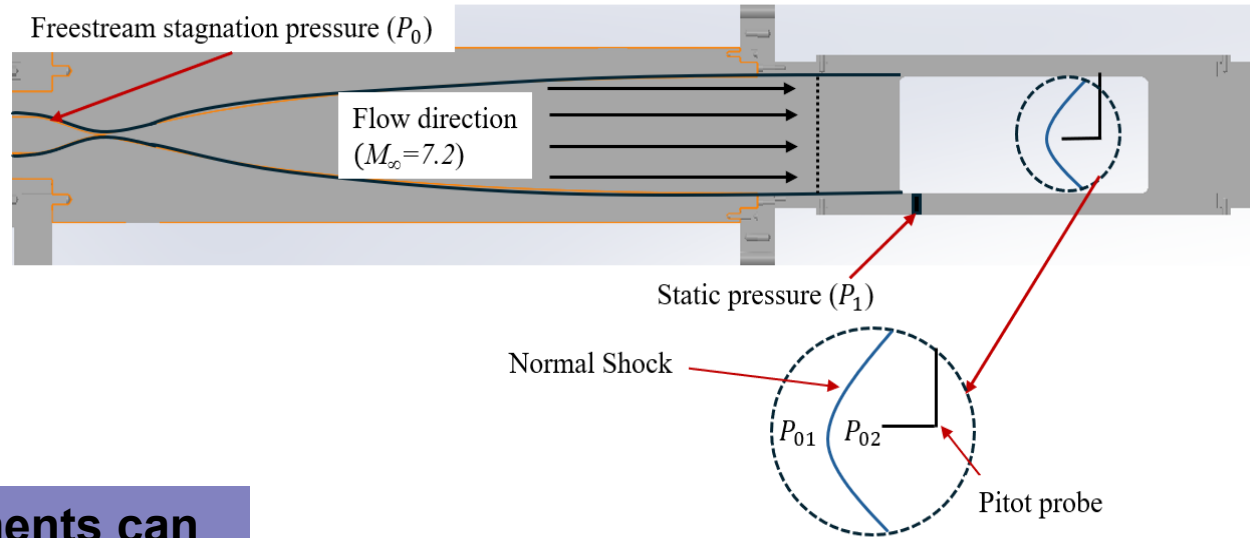
UTSA Mach 7 Ludwieg Tube

- Nominal Mach number: 7.2
- High-pressure driver & low-pressure vacuum setup
- Diaphragm rupture initiates flow
- Flow expands through the nozzle to the test section
- Steady state passes (70-100 ms)



Pitot Probe Measurements

- Freestream stagnation pressure P_0
- Static pressure port $\rightarrow P_1$
- Normal shock at Pitot probe $\rightarrow P_{02}$



Pitot probe measurements can reveal how flow properties vary across the test section.

Freestream Mach Number

$$\frac{P_{02}}{P_0} = \left(\frac{\frac{\gamma + 1}{2} M_1^2}{1 + \frac{\gamma - 1}{2} M_1^2} \right)^{\frac{\gamma}{\gamma - 1}} \left(\frac{1}{\frac{2\gamma}{\gamma + 1} M_1^2 - \frac{\gamma - 1}{\gamma + 1}} \right)^{\frac{1}{\gamma - 1}}$$

- Total pressure ratio $P_{02}/P_0 \rightarrow$ determines Mach number
- Freestream stagnation pressure $P_0 = P_{01}$

Static Pressure Estimate

$$\frac{P_1}{P_0} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{\gamma}{\gamma - 1}}$$

- Static pressure difficult to measure reliably in high-speed flow
- Estimate P_1 using isentropic relation (P_1/P_0)
- Assume P_1 constant across boundary layer

Rayleigh Pitot Method

$$\frac{P_{02}}{P_1} = \left[\frac{(\gamma+1)^2 M^2}{4\gamma M^2 - 2(\gamma-1)} \right]^{\frac{1}{\gamma-1}} \left[\frac{1-\gamma+2\gamma M^2}{\gamma+1} \right]^{\frac{\gamma}{\gamma-1}}$$

- Rayleigh Pitot relation → Mach number
- Uses P_{02} (measured) and P_1 (estimated)

Static Pressure Uncertainty

$$\delta P_1 = \sqrt{\left(\frac{\partial P_1}{\partial P_{02}} \delta P_{02}\right)^2 + \left(\frac{\partial P_1}{\partial P_0} \delta P_0\right)^2}$$

- Uncertainty in P_1 estimated using Kline–McClintock method
- Uses uncertainties in P_0 and P_{02} (manufacturer data)

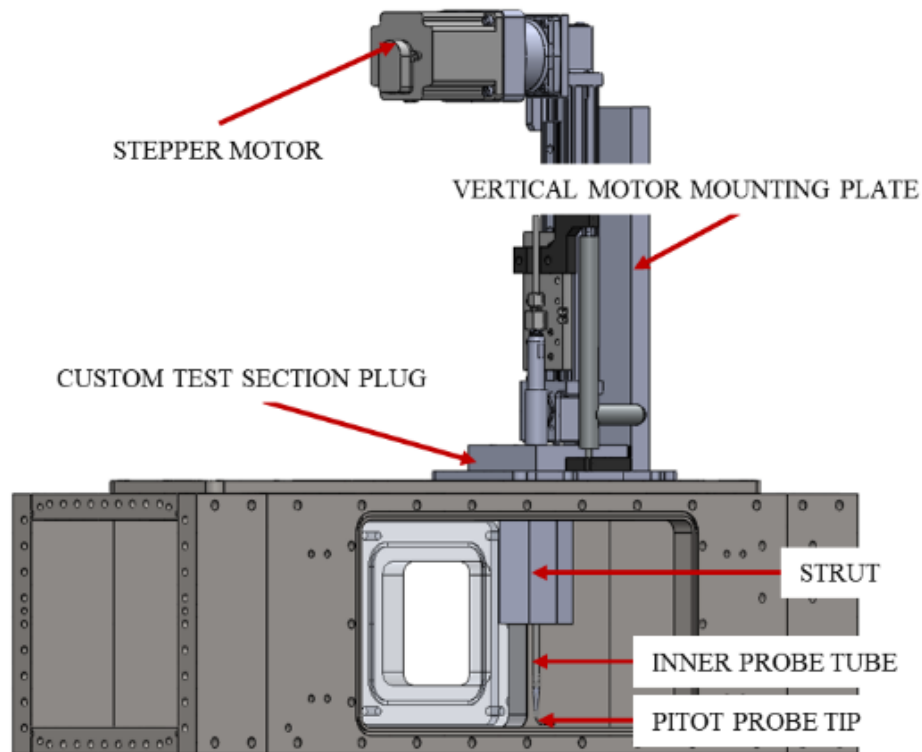
Mach Number Uncertainty

$$\delta M = \sqrt{\left(\frac{\partial M}{\partial P_{02}} \delta P_{02}\right)^2 + \left(\frac{\partial M}{\partial P_1} \delta P_1\right)^2}$$

- Mach depends on pressure ratio (P_{02}/P_1)
- Uncertainty comes from both P_{02} and estimated P_1
- Kline–McClintock method used for Mach uncertainty
- Mach is highly sensitive to changes in P_1
- Small changes in P_1 cause large variations in Mach number

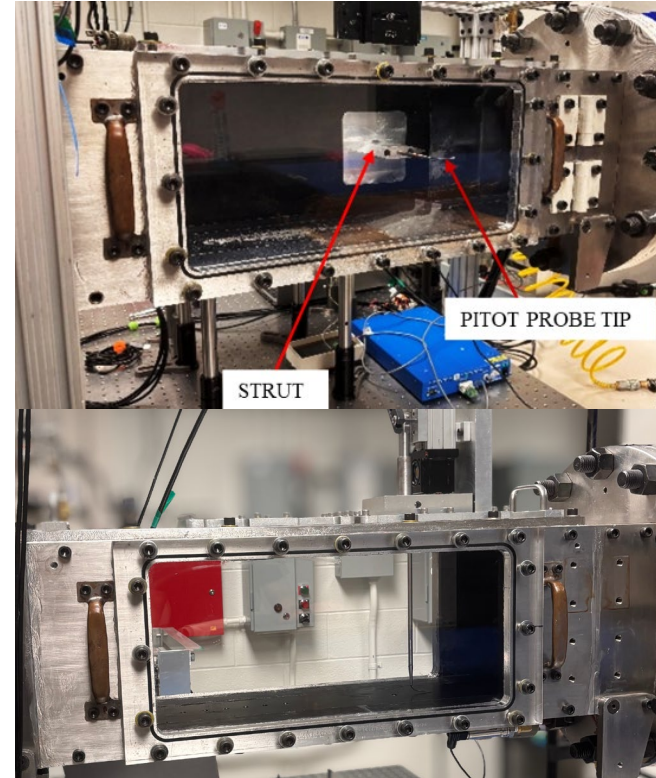
Probe System Overview

- Pitot Probe mounted using a Stepper motor Traversing system
- Laser used to align the Pitot probe in the center of the flow
- Strut implemented to prevent oscillations



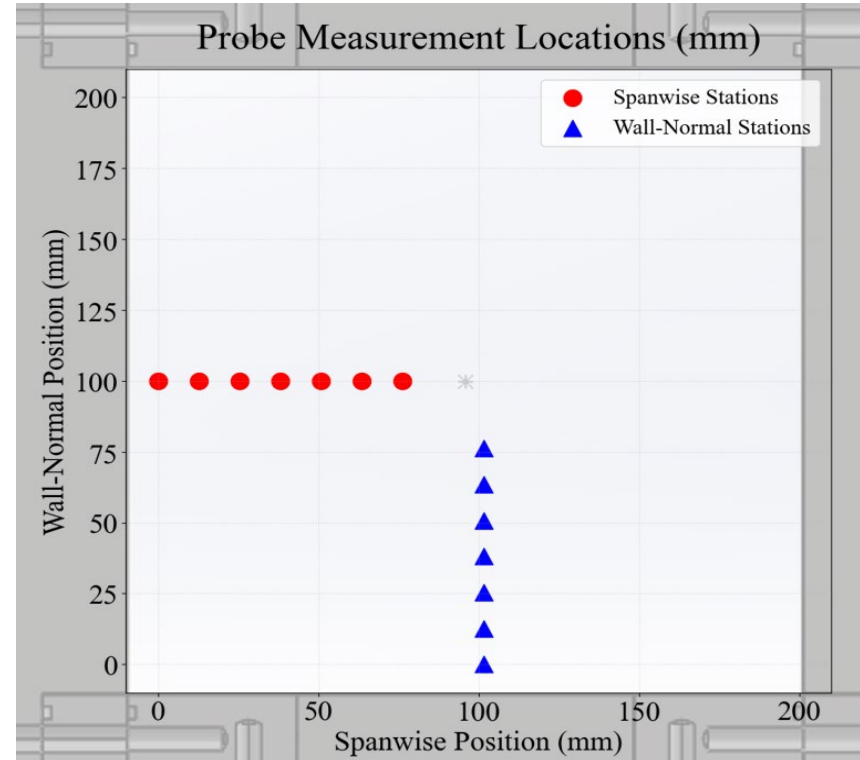
Experimental Setup

- Probe was traversed in both spanwise and wall-normal directions
- Motor mount configuration allows positioning from side wall or top
- Measurements were taken pointwise from the wall toward the freestream



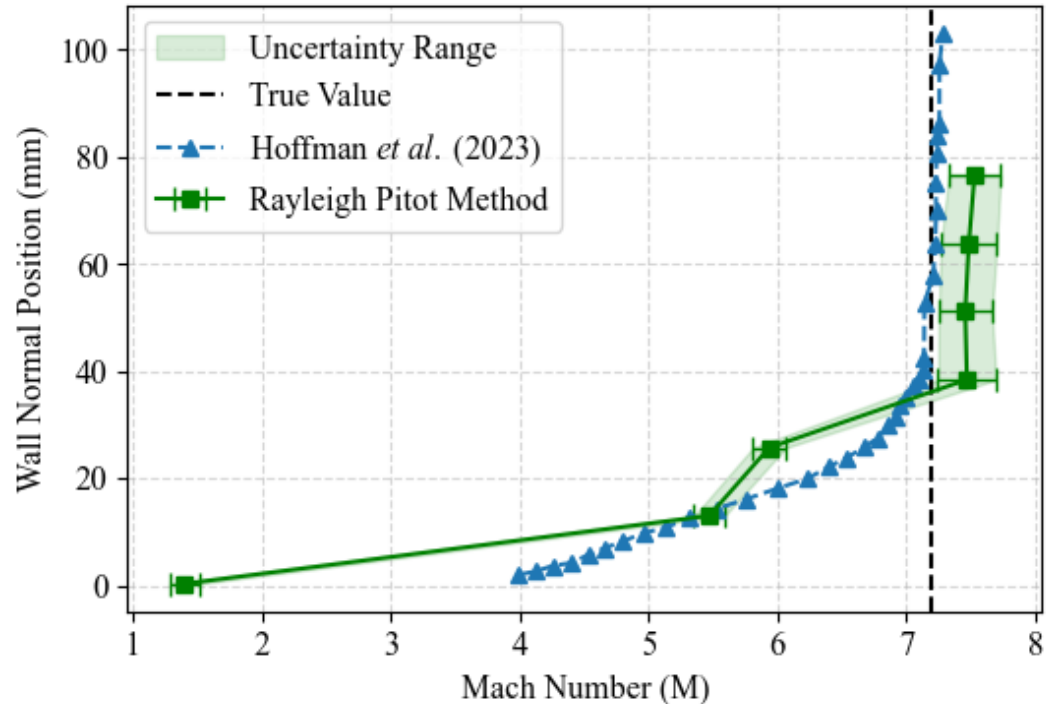
Test Section Measurement Locations

- Total of fourteen runs: seven wall-normal and seven spanwise
- Seven discrete points spaced 12.75 mm apart
- Points extend from wall into the flow for both directions



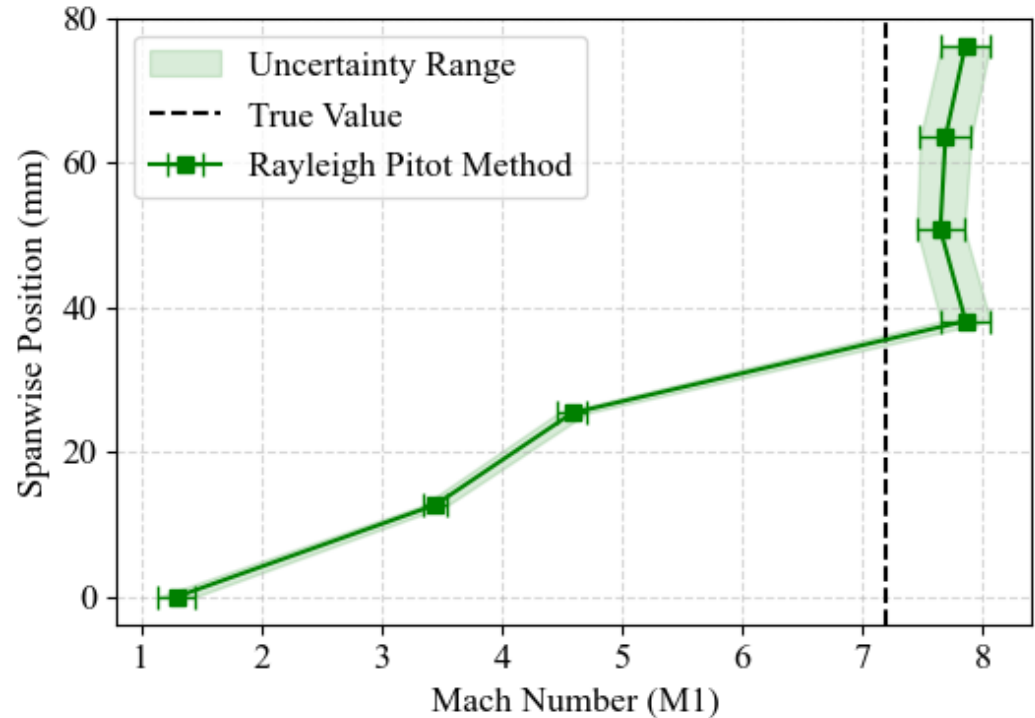
Wall-Normal Results

- Profile shows agreement with initial tunnel characterization scans from (Hoffman *et al.* 2023)
- Freestream Mach ≈ 7.2 within uncertainty bounds
- Boundary layer thickness ≈ 40 mm
- Differences near floor due to pointwise measurements



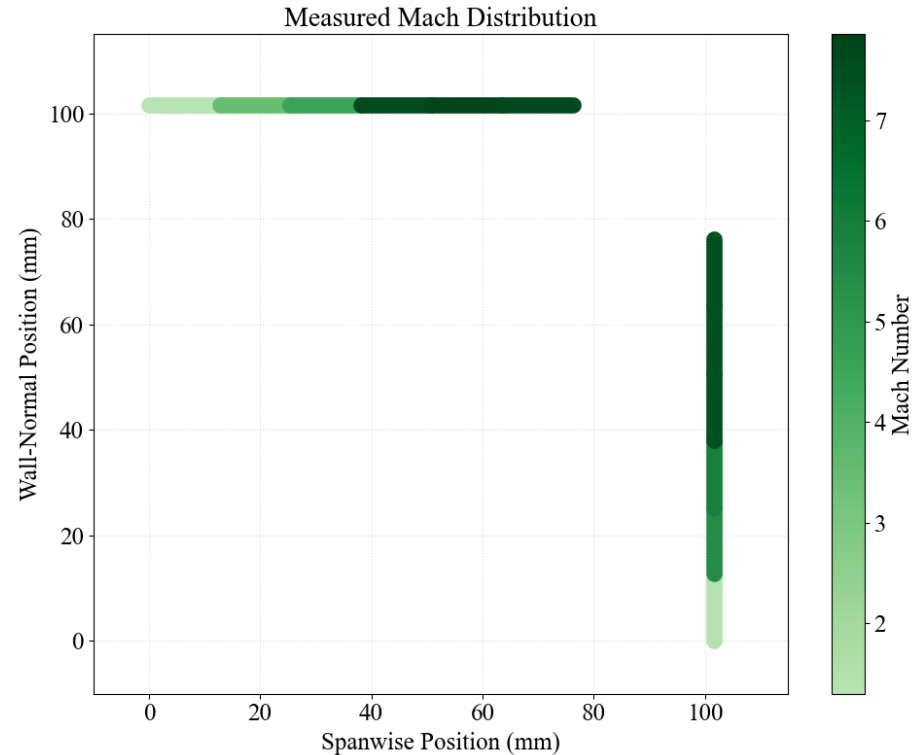
Spanwise Results

- Boundary layer thickness estimated at ~40 mm
- Freestream Mach number agrees with wall-normal profile



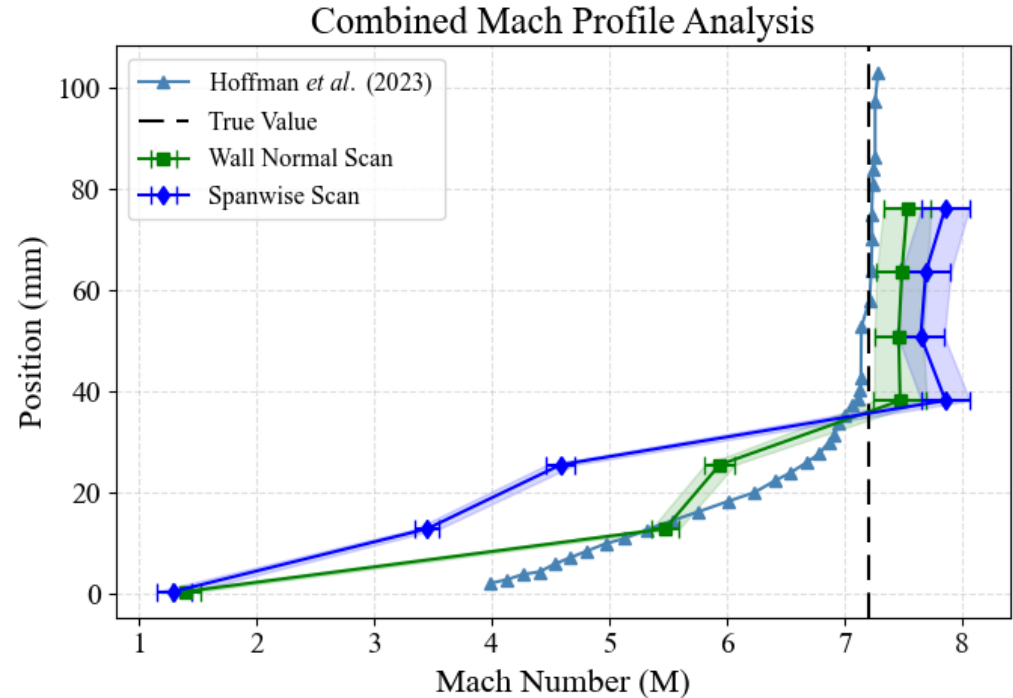
Mach Field Reconstruction

- Spatial Mach number distribution of both the wall-normal and spanwise
- Uniform freestream region validated
- One quadrant shown - upper right of the test section center



Comparison of Mach Profiles

- Combined wall-normal and spanwise profiles with prior data
- Freestream Mach agrees with characterization (≈ 7.2)
- Overall profile shape shows strong agreement



Conclusion

- Wall-normal measurements captured boundary layer development
- Spanwise measurements show reduced Mach near sidewall boundary layer
- Freestream Mach number consistent with prior facility characterization
- Results confirm stable Mach 7.2 operation of the UTSA Ludwig tube



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